Rat Detection AI

Simple Results Report

RESULT: The model works and is ready to use.
It finds about 8 out of 10 rats, is tiny (57 KB), and runs on the ESP32 camera.

78%	57 KB	0.70
of rats caught	**model size**	**detection score**
at the normal setting	fits on the camera	F1, higher is better

In one line: a small AI that watches the feeding station and tells you when a rat is there.

Two words used in this report:

Recall = out of all the real rats, how many did it catch? (like a fishing net)

Precision = when it shouts “rat!”, how often is it right? (not crying wolf)

1. What is this, and how does it work?

This is a tiny AI “brain” for your camera. It looks at the feeding station and answers one simple question: is there a rat in view right now?

How it works, in simple steps:

1. The camera takes a small black-and-white photo (96 x 96 pixels).
2. The AI splits the photo into a 12 x 12 grid, like a checkerboard.
3. For each square it gives a score: how likely a rat’s centre is there.
4. Squares with a high score are grouped together — each group is one rat.

It does not draw exact boxes around rats; it just drops a “dot” where each rat is. That is why it can be so small and fast — small enough to run on a cheap camera board.

2. What it learned from

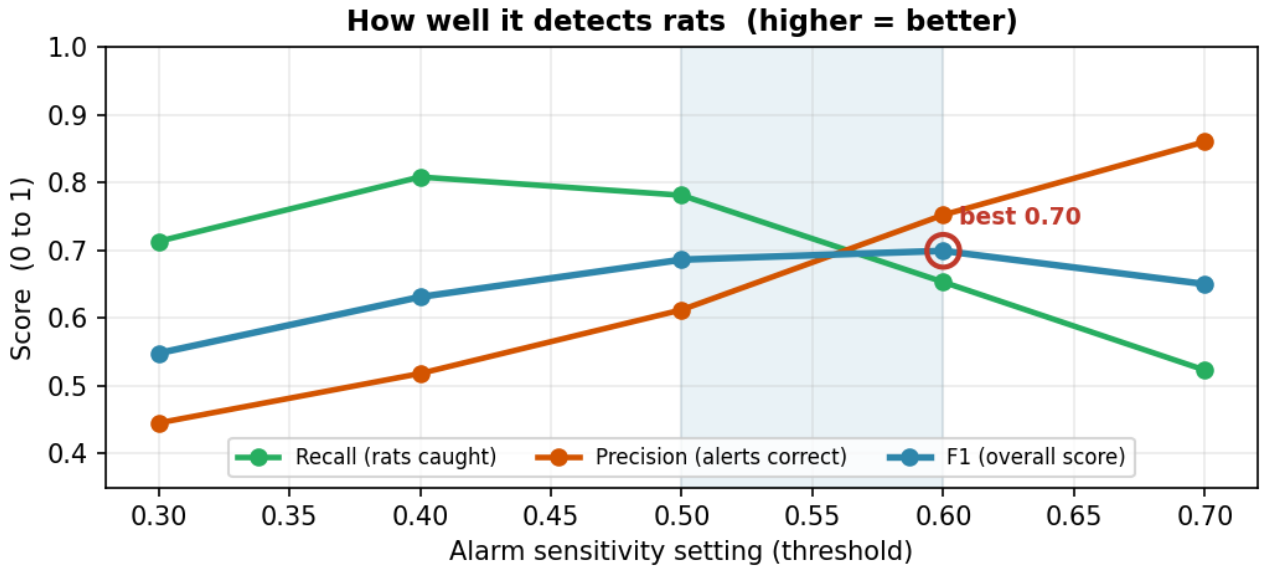
The AI was shown thousands of pictures. Some had rats, some were empty (so it also learns what “no rat” looks like).

Pictures	Total	With rats	Empty
To learn (training)	4,192	3,600	592
To test (the exam)	1,040	900	140

Important: the test pictures are kept separate, so the “exam” score below is honest — the AI never saw those pictures while learning.

3. How well does it work?

There is always a trade-off. One setting (the “threshold”) decides how eager the AI is to shout “rat!”. Turn it up and you get fewer false alarms but miss more rats; turn it down and you catch more rats but get more false alarms.



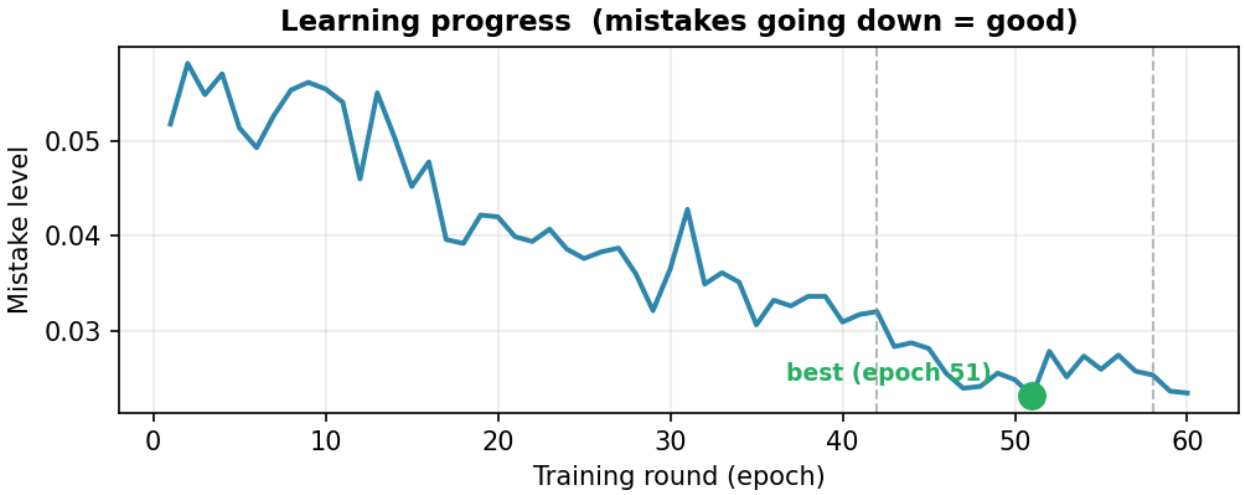
Two sensible settings to choose from:

Setting	Rats caught	Alerts correct	Best for
Normal (0.5)	78%	61%	catching every rat
Careful (0.6)	65%	75%	fewer false alarms

The camera currently uses the Normal (0.5) setting.

4. Did it learn properly?

“Training” means the AI reviewed the pictures 60 times to improve. The graph below shows its mistakes going down smoothly and staying down — that is exactly what a healthy, well-behaved training looks like (no sudden jumps, no breakdowns).



5. Conclusion

- The model is good for its job: a rat alarm.
- Strength: it rarely misses rats (it catches about 8 out of 10).
- Weak point: sometimes it alarms when there is no rat (false alarms).
- Easy fixes for false alarms: (a) make the camera wait until a rat shows up in several photos in a row, or (b) use the “Careful” setting.
- Next step: put it on the camera and test it in real life.

Technical note: FOMO detector (MobileNetV2 alpha=0.35), input 96x96 grayscale, output 12x12 grid, ~20.7K parameters, exported int8 TFLite (57 KB). Best object-level F1 = 0.699 at threshold 0.6; recall 0.78 at threshold 0.5.